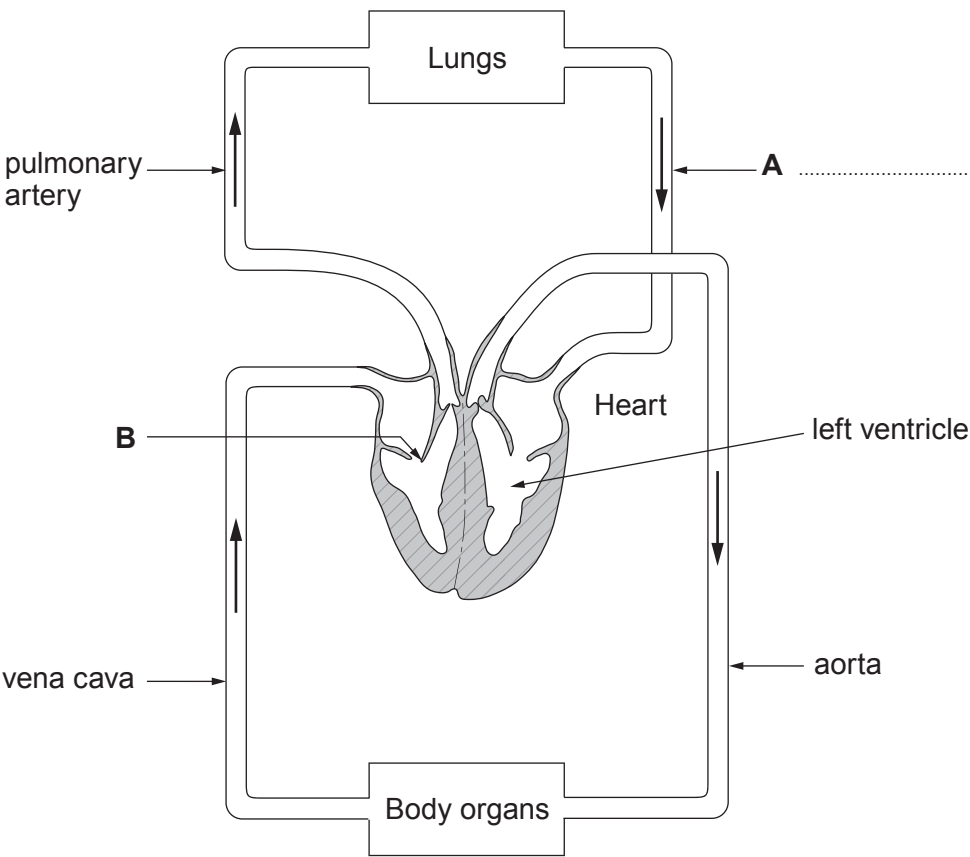


3. The diagram shows the human circulatory system.



- (a) (i) Label the blood vessel **A** on the diagram. [1]
- (ii) State the function of structure **B**. [1]
-



(b) The table shows blood pressures in different parts of the circulatory system.

Blood vessel	Maximum blood pressure (kPa)
pulmonary artery	3.3
aorta	16.0
capillary in body organ	2.0
left ventricle	17.0
right ventricle	3.5

- (i) I. Calculate the difference in the maximum blood pressure between the aorta and the pulmonary artery. [1]

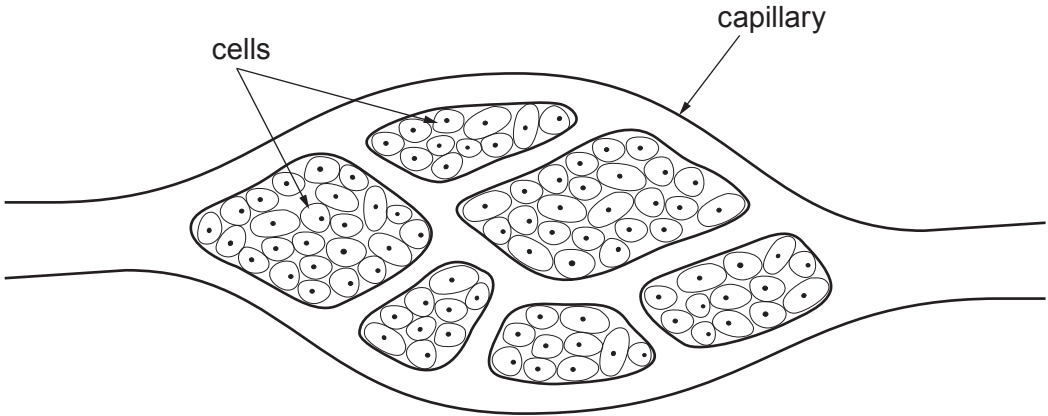
Difference in pressure = kPa

- II. State the reason for the difference in blood pressure in these two blood vessels. [1]

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.....

- (ii) The diagram shows some capillaries in an organ of the body.



Blood flows very slowly through the capillaries, allowing useful substances in the blood and waste products in cells to be exchanged.

- I. From the table above, what is the evidence that the blood flows slowly through the capillaries? [1]

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- II. State how the capillary walls are well adapted for the exchange of substances. [1]

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